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# BAR: Budgeted Actor Refinement Decouples Bootstrap Exposure from Policy Reach

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## Abstract

TD3+BC uses the same actor branch for bootstrapped critic targets and deployment, coupling bootstrap exposure to policy reach. We introduce budgeted actor refinement (BAR), a persistent  $K$ -actor chain whose first actor supplies TD-target actions and whose final actor is deployed. A finite-step bound makes target perturbation depend on first-hop displacement, while an ideal Wasserstein-2 proximal analysis describes the complementary local geometry. On nine D4RL locomotion tasks, fourteen budgets, and four proximal seeds, the equally weighted  $T \geq 4$  mean is strictly ordered by depth, rising from 25.38 at  $K = 1$  to 63.97 at  $K = 4$  as score- $< 20$  cells fall from 34/63 to 8/63; both host-separated seed pairs reproduce the full ordering. A four-seed projected-linearized sweep also gains 8.01 points from  $K = 2$  to  $K = 3$ , with a 95% task-resampling interval [3.56, 13.53]. The proximal  $K = 4 - K = 3$  gain is 12.06 with a 95% interval [3.84, 21.11], but its .34 cellwise median and eight rescues show that lower-tail recovery drives the aggregate. A matched 270-checkpoint TD3/P2/P3 audit finds that BAR-Prox  $K = 3$  lowers next-state target displacement relative to TD3+BC in 88/90 cells while typical final-policy displacement does not contract; a separate four-seed  $K = 4$  audit lowers target displacement in 89/90 matched seed-0/1 comparisons with TD3+BC. Thus depth is a finite-step stability control: moderate refinement decouples exposure from reach, while added depth further suppresses first-hop exposure and res-

cues failures.

## 1 INTRODUCTION

Offline reinforcement learning (RL) seeks a policy from a fixed dataset without additional environment interaction. Its central difficulty is not merely to select a high-value action, but to determine how far policy improvement can proceed before the learned critic becomes unreliable. TD3+BC regularizes a deterministic actor toward dataset actions while using a Polyak copy of that actor in bootstrapped critic targets (Fujimoto and Gu, 2021). The same policy branch therefore determines both the action shift entering critic training and the reach of the evaluated policy. Aggressive improvement can move target actions into regions where critic error is amplified, whereas conservative improvement can leave useful policy improvement unrealized.

We study this coupling through *budgeted actor refinement* (BAR), a re-centered actor architecture for TD3+BC. BAR represents a nominal improvement budget with  $K$  sequential actor updates. The first actor is anchored to dataset actions, and each later actor is anchored to the stop-gradient output of its predecessor. A target copy of the first refined actor supplies the next actions in TD targets, while the final actor is used for evaluation. The architecture thus assigns critic-target construction and deployment to distinct policy branches without changing the critic update rule.

Let  $D_{\text{critic}}$  denote the squared displacement of the critic-coupled first-hop branch and let  $D_{\text{final}}$  denote that of the evaluated final policy. A one-hop actor identifies these quantities; BAR assigns them to  $\mu_1$  and  $\mu_K$ , respectively. This separation can increase final-policy reach at comparable first-hop movement or contract both movements near an unstable training outcome. It also introduces a countervailing mismatch because the critic target follows the  $\mu_1$  branch while evaluation follows  $\mu_K$ . The method therefore exposes a design axis between target-policy displacement and

deployment reach rather than guaranteeing that either quantity is independently optimal.

For deterministic actors, the squared TD3+BC anchor is a Wasserstein-2 movement cost in the Dirac-policy limit. With a fixed critic and small realized steps, ideal explicit and implicit updates discretize the same flow to first order. The large-budget tail instead probes the coupled finite-step regime, where projection, actor fitting, critic drift, saturation, and optimization history determine the end-to-end outcome.

The complete proximal phase diagram yields high-budget means  $25.38 \rightarrow 41.20 \rightarrow 51.92 \rightarrow 63.97$  for  $K = 1, 2, 3, 4$  and low-return counts  $34 \rightarrow 22 \rightarrow 16 \rightarrow 8$ . Both disjoint seed pairs reproduce the ordering; eight rescues account for 54.7% of the  $K = 3 \rightarrow 4$  gain. A four-seed projected-linearized sweep reproduces the  $K = 2 \rightarrow 3$  direction. Direct diagnostics separate a moderate-depth regime, where target exposure contracts without typical final-policy contraction, from added-depth first-hop contraction. Failed runs also exhibit extreme TD-error tails, larger movement, and gain reversal under a common critic.

Our contributions are as follows.

- We formulate BAR for TD3+BC as a re-centered actor composition that assigns target-policy displacement and final-policy reach to distinct branches.
- We give a Wasserstein proximal interpretation, derive an inexact finite-step margin with an explicit critic-error term, bound first-hop TD-target perturbation, and separate these statements from the fixed-critic small-step flow limit.
- Four-seed proximal sweeps through  $K = 4$ , a four-seed linearized sweep through  $K = 3$ , direct target-action measurements, and targeted controls show that depth broadens the large-budget stability envelope; the largest proximal gain is driven primarily by lower-tail rescue.

## 2 RELATED WORK

**Behavior-regularized offline actor-critic learning.** Offline actor-critic methods must control value extrapolation when a learned policy selects actions outside the fixed dataset. Prior work studies this failure through constrained policy improvement, conservative value learning, and restrictions on actions used in Bellman backups (Fujimoto et al., 2019; Kumar et al., 2019). TD3+BC provides a particularly minimal control: a deterministic actor maximizes a normalized critic while a squared cloning term anchors its

actions to the dataset (Fujimoto and Gu, 2021). We keep this critic update and actor family fixed and study which refined actor branch supplies the TD target. One-step offline RL instead limits repeated policy optimization against an off-policy value estimate (Brandfonbrener et al., 2021). BAR retains iterative actor-critic training but distributes a nominal improvement coefficient across a routed actor chain.

### Policy geometry and proximal refinement.

Trust-region policy optimization is commonly derived from information geometry (Amari and Nagaoka, 2000; Schulman et al., 2015). Optimal-transport geometry gives a different notion of movement: policy optimization can be expressed as a Wasserstein gradient flow, with related natural-gradient and trust-region constructions for continuous control (Zhang et al., 2018; Moskovitz et al., 2020; Terpin et al., 2022; Song et al., 2023). Wasserstein Policy Optimization projects Wasserstein policy flow into a parametric actor class using value-function action gradients (Pfau et al., 2025). For deterministic policies, the statewise Wasserstein-2 cost reduces to squared action displacement. We use that identity to interpret the TD3+BC anchor as a proximal movement cost, while separating the finite-step variational statement from its small-step flow limit.

### Target networks and actor-critic coupling.

Bootstrapping error can feed critic extrapolation back into actor updates (Fujimoto et al., 2019; Kumar et al., 2019). Polyak averaging delays this feedback but ordinarily copies the same actor branch that will be deployed. MCEP instead pairs a conservative target policy for value learning with a mildly constrained evaluation policy excluded from Bellman backups (Xu et al., 2024). BAR shares this routing principle but uses a persistent, predecessor-recentered chain and varies depth under a fixed nominal budget; MCEP keeps both actors directly data-anchored with separately chosen strengths and is not a  $K = 2$  instance of BAR. Our contribution concerns this depth-indexed routing operator rather than a new critic objective.

## 3 PRELIMINARIES

We consider a discounted Markov decision process  $\langle \mathcal{S}, \mathcal{A}, P, r, \gamma \rangle$  with continuous actions and a fixed offline dataset  $\mathcal{D} = \{(s_i, a_i, r_i, s'_i)\}_{i=1}^N$ . A clipped double critic is trained from targets

$$y_\rho = r + \gamma \min_j Q_j^-(s', \rho(s')), \quad (1)$$

where  $\rho$  is a target policy. When  $\rho(s')$  leaves dataset support, critic error enters the regression target and

can be amplified by later actor updates (Fujimoto et al., 2019; Kumar et al., 2019).

For a deterministic actor  $\mu_\theta$ , normalized TD3+BC minimizes

$$\mathcal{L}_{\text{actor}}(\theta) = -\alpha \mathbb{E}_s \bar{Q}(s, \mu_\theta(s)) + \mathbb{E}_{(s,a) \sim \mathcal{D}} \frac{\|\mu_\theta(s) - a\|_2^2}{d}, \quad (2)$$

where  $\bar{Q} = \hat{Q}/C$  and  $C > 0$  is a stop-gradient critic scale (Fujimoto and Gu, 2021). Ordinarily, a Polyak copy of this actor enters Eq. (1), while the online actor is deployed. Target-policy movement and deployment reach therefore follow the same branch.

A metric proximal update with step  $h > 0$  takes the form

$$\pi^+ \in \arg \min_{\pi} \left\{ \hat{\mathcal{E}}(\pi) + \frac{1}{2h} d_{\mathcal{M}}^2(\pi, \pi^{\text{ref}}) \right\}. \quad (3)$$

This minimizing-movement construction is associated with metric gradient flows (Ambrosio et al., 2005; Otto, 2001). For deterministic state-conditioned policies,

$$W_2^2(\delta_{\mu(s)}, \delta_{\nu(s)}) = \|\mu(s) - \nu(s)\|_2^2. \quad (4)$$

Thus the TD3+BC anchor is a Wasserstein-2 movement cost up to fixed scaling. We distinguish its finite-step variational meaning from the small-step, fixed-critic gradient-flow limit and from large-budget coupled training.

## 4 BUDGETED ACTOR REFINEMENT

At each delayed actor update, BAR maintains  $K$  independently initialized, persistent deterministic actors with separate optimizer states. The first actor takes one TD3+BC update toward the sampled dataset actions; each later actor takes one update toward the stop-gradient output of its predecessor on the same minibatch. The online critic is fixed within this chain; we denote its first Q-function by  $\hat{Q}$ . A Polyak copy of the first actor supplies TD-target actions, and the final actor is evaluated. We first derive the proximal operator underlying these updates and then formalize their composition.

### 4.1 Offline actor updates as proximal improvement

Let  $\rho_{\mathcal{D}}$  be the empirical state distribution and consider a deterministic actor  $\mu : \mathcal{S} \rightarrow \mathcal{A} \subset \mathbb{R}^d$ . At one training iteration, the critic parameters are held fixed while the actor chain is updated. We write the current critic as  $\hat{Q}_t$ , use  $\epsilon = 10^{-6}$ , and define the implementation-level

proximal scales by

$$\begin{aligned} C_{1,t}^{\text{P}} &= \text{sg} \left( \mathbb{E}_s |\hat{Q}_t(s, \mu_{1,t}^{\text{pre}}(s))| + \epsilon \right), \\ C_{k,t}^{\text{P}} &= \text{sg} \left( \mathbb{E}_s |\hat{Q}_t(s, \text{sg } \mu_{k-1,t}^+(s))| + \epsilon \right), \quad k \geq 2. \end{aligned} \quad (5)$$

The normalized critic is  $\bar{Q}_{k,t}(s, a) = \hat{Q}_t(s, a)/C_{k,t}^{\text{P}}$ . Here *pre* and the superscript  $+$  denote actions before and after the current actor update, and *sg* denotes stop-gradient. We suppress  $t$  and the superscript when discussing the ideal operator. A fixed positive  $C$  only rescales gradient-flow time; recomputing  $C_k$  changes the finite-step schedule. Our finite-step statements condition on the realized scales.

For deterministic policies and a state distribution  $\rho$ , define the action-averaged Wasserstein metric

$$\mathbf{d}_{d,\rho}^2(\mu, \nu) := \mathbb{E}_{s \sim \rho} \left[ \frac{1}{d} \|\mu(s) - \nu(s)\|_2^2 \right]. \quad (6)$$

It is  $d^{-1}$  times the state-averaged squared  $W_2$  distance between state-conditioned Dirac policies. We abbreviate  $\mathbf{d}_d = \mathbf{d}_{d,\rho_{\mathcal{D}}}$  on the actor-training state distribution. The population counterpart of squared behavior cloning also admits a deterministic reference. Let  $P_B$  be the data-generating behavior distribution, let  $\rho_B$  be its state marginal, and define  $b(s) = \mathbb{E}_{P_B}[a | s]$ . Then

$$\mathbb{E}_{(s,a) \sim P_B} \frac{1}{d} \|\mu(s) - a\|_2^2 = \mathbf{d}_{d,\rho_B}^2(\mu, b) + \text{const}, \quad (7)$$

where the constant is the conditional action variance and does not depend on  $\mu$ . The empirical actor loss is a Monte Carlo estimate of the left-hand side, so samplewise squared anchoring targets the conditional-mean behavior map at the population level.

Given a reference policy  $\nu$ , step  $h > 0$ , and fixed normalized critic  $\bar{Q}$ , define the proximal refinement operator by

$$\mathcal{P}_h^{\bar{Q}}(\nu) \in \arg \min_{\mu \in \mathcal{M}} \left\{ -\mathbb{E}_s \bar{Q}(s, \mu(s)) + \frac{1}{2h} \mathbf{d}_d^2(\mu, \nu) \right\}. \quad (8)$$

At hop  $k$ , setting  $\bar{Q} = \bar{Q}_k$  and multiplying by  $2h$  gives the practical actor objective

$$\mathcal{L}_{h,k}(\mu | \nu) = -2h \mathbb{E}_s \bar{Q}_k(s, \mu(s)) + \mathbf{d}_d^2(\mu, \nu). \quad (9)$$

**Proposition 1** (Behavior anchoring is Wasserstein proximal improvement). *For deterministic policies, the normalized anchored objective*

$$-\alpha \mathbb{E}_s \bar{Q}(s, \mu(s)) + \mathbf{d}_d^2(\mu, \nu)$$

*is Eq. (8), up to multiplication by a positive constant, with  $h = \alpha/2$ . Equivalently, the raw-critic objective  $-\lambda \mathbb{E}_s \hat{Q}(s, \mu(s)) + \mathbf{d}_d^2(\mu, \nu)$  has  $h = \lambda C/2$  when  $C$  is fixed. The population counterpart of squared dataset anchoring uses  $\nu = b$  by Eq. (7).*

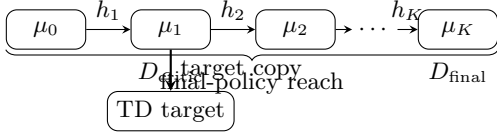


Figure 1: BAR re-centers each actor update. A Polyak copy of  $\mu_1$  supplies TD-target actions, while later actors increase or reshape final-policy reach without entering the target policy.

The proposition identifies an update operator relative to the *estimated* critic; true-return improvement lies outside the scope of this certificate.

#### 4.2 Re-centered composition and actor-critic dataflow

For population analysis, let  $\mu_0 := b$  denote the conditional-mean behavior map and let  $h_{1:K}$  be positive step sizes. The idealized BAR chain is

$$\mu_k \in \mathcal{P}_{h_k}^{\bar{Q}_k}(\text{sg } \mu_{k-1}), \quad k = 1, \dots, K. \quad (10)$$

The implementation approximates each subproblem with one Adam update. Its first actor  $\mu_1$  uses sampled dataset actions as its reference; later actors  $\mu_2, \dots, \mu_K$  use the preceding online actor under the same current critic. Only the Polyak copy of  $\mu_1$  enters the TD target, while  $\mu_K$  supplies the evaluated action. Because the later persistent actors are independently initialized, their anchoring gradient need not vanish even at  $h = 0$  when predecessor and current actor differ. The ideal proximal map therefore motivates, but does not certify, the live one-Adam-step chain.

Our fixed-budget schedule uses the implementation parameter  $T = \tau = \alpha/2$ , sets  $h_k = T/K$ , and holds  $T$  fixed. Thus  $K = 1$  recovers the TD3+BC coefficient  $\alpha = 2T$ ;  $T$  is neither a compute budget nor a realized distance constraint. Increasing  $K$  jointly changes local coefficient, re-centering frequency, actor count and compute, normalization trajectory, and target/deployment dataflow. Each  $K$  is therefore a complete randomized procedure, and its seed average estimates marginal end-to-end performance. Identical critic initialization across depths is a common-random-number variance-reduction design, not a validity condition.

On the next-state distribution  $\rho'_{\mathcal{D}}$  used by the TD target, define the action-averaged population displacements

$$\begin{aligned} D_{\text{critic}} &:= d_{d, \rho'_{\mathcal{D}}}^2(\mu_1, \mu_0), \\ D_{\text{final}} &:= d_{d, \rho'_{\mathcal{D}}}^2(\mu_K, \mu_0). \end{aligned} \quad (11)$$

A one-step actor has  $D_{\text{critic}} = D_{\text{final}}$ ; BAR assigns the two population quantities to distinct policy branches.

On fixed validation batches, we measure the sample-anchored online branches and the deterministic next-state Polyak target branch:

$$\begin{aligned} \hat{D}_{\text{critic}} &= \mathbb{E}_i \|\mu_1(s_i) - a_i\|_2^2/d, \\ \hat{D}_{\text{final}} &= \mathbb{E}_i \|\mu_K(s_i) - a_i\|_2^2/d, \\ \hat{D}_{\text{target}} &= \mathbb{E}_i \|\mu_1^-(s'_i) - a_{i+1}\|_2^2/d. \end{aligned} \quad (12)$$

The direct audit retains only bootstrapped transition pairs; a smoothed version uses common TD3 target-noise draws. Sample anchoring adds a common conditional-action-variance floor in population differences. These are movement measurements, not critic-reliability estimates.

#### 4.3 Finite-step statements

The following statements do not require a small-step Taylor expansion.

**Proposition 2** (Inexact estimated-energy margin). *Suppose  $\mu_{k-1}$  is feasible for the  $k$ th actor class and the computed actor satisfies*

$$\mathcal{L}_{h_k, k}(\mu_k \mid \mu_{k-1}) \leq \mathcal{L}_{h_k, k}(\mu_{k-1} \mid \mu_{k-1}) + \varepsilon_k.$$

Then

$$\begin{aligned} m_k^{\hat{Q}} &:= \frac{2h_k}{C_k} \mathbb{E}_s [\hat{Q}(s, \mu_k(s)) - \hat{Q}(s, \mu_{k-1}(s))] \\ &\quad - d_d^2(\mu_k, \mu_{k-1}) \geq -\varepsilon_k. \end{aligned} \quad (13)$$

For the first sample-anchored hop, a statewise dataset action is not generally a feasible deterministic-policy comparator; the population reference in Eq. (7) or a behavior actor is required for a literal certificate. For re-centered hops, the preceding actor is feasible by construction.

The margin is relative to  $\hat{Q}$ . Let a trusted objective be  $Q^\dagger$  and write  $\hat{Q} = Q^\dagger + e$ . Then

$$\begin{aligned} m_k^{Q^\dagger} &= m_k^{\hat{Q}} - \frac{2h_k}{C_k} \mathbb{E}_s [e(s, \mu_k(s)) \\ &\quad - e(s, \mu_{k-1}(s))]. \end{aligned} \quad (14)$$

Thus estimated-energy descent transfers to  $Q^\dagger$  only when critic error is controlled along the realized path. Equation (13) certifies the learned objective rather than monotone true return.

**Proposition 3** (First-hop TD-target perturbation bound). *For clarity, consider a noiseless TD target using  $\mu_1$ ,  $y_{\mu_1} = r + \gamma \min_j Q_j^-(s', \mu_1(s'))$ , and suppose each target critic  $Q_j^-(s', \cdot)$  is  $L(s')$ -Lipschitz. For any reference  $\mu_0$ ,*

$$\begin{aligned} \|y_{\mu_1} - y_{\mu_0}\|_{L_2(\mathcal{D})} &\leq \gamma \left( \mathbb{E}_{s'} L(s')^2 \right. \\ &\quad \cdot \left. \|\mu_1(s') - \mu_0(s')\|_2^2 \right)^{1/2}. \end{aligned} \quad (15)$$

If  $L(s') \leq \bar{L}$ , the right-hand side is at most  $\gamma \bar{L} \sqrt{dD_{\text{critic}}}$ . Conditional on  $\mu_1$ , later actors do not enter this one-step target perturbation.

Proposition 3 bounds a one-step value-target difference, not the stability of the learned critic. Its architectural content is that later refinement can alter  $D_{\text{final}}$  without directly entering this bound. With a Polyak target network,  $\mu_1$  is replaced by its target copy. A same-noise comparison under TD3 target smoothing obeys the same bound because action clipping is non-expansive. The current-state  $\hat{D}_{\text{critic}}$  omits Polyak lag and target noise, whereas  $\hat{D}_{\text{target}}$  measures the Polyak actor directly and has a smoothed counterpart. Neither displacement measures the action Lipschitz factor or the resulting change in critic value.

The separation introduces a complementary evaluation-policy mismatch: critic targets follow the  $\mu_1$  branch, whereas evaluation follows  $\mu_K$ . Increasing  $D_{\text{final}}$  at fixed  $D_{\text{critic}}$  can therefore make the critic less representative of the evaluated policy. We interpret BAR as exposing a measurable target/deployment trade-off; its empirical return captures the net effect of target movement, evaluation mismatch, actor compute, and the coupled optimization trajectory.

#### 4.4 Local Wasserstein-flow limit

The gradient-flow interpretation is a local organizing principle. Fix a critic and an interior action  $a_0$ . Let  $g = \nabla_a \hat{Q}(a_0)$ ,  $H = \nabla_a^2 \hat{Q}(a_0)$ , and  $\beta = dh/C$ . Before projection, the action-metric-matched linearized and exact statewise proximal updates satisfy

$$\Delta a_{\text{lin}} = \beta g, \quad \Delta a_{\text{prox}} = \beta \nabla_a \hat{Q}(a_0 + \Delta a_{\text{prox}}). \quad (16)$$

If the Hessian is locally Lipschitz and  $\beta \|H\| < 1$ , the latter yields

$$\Delta a_{\text{prox}} = \beta g + \beta^2 H g + O(\beta^3), \quad (17)$$

and both ideal updates are first-order discretizations of  $\dot{a} = d\nabla_a \hat{Q}(a)$ . With fixed  $C$  this is a constant time-rescaling of raw-critic gradient ascent; with a positive policy-dependent global  $C$  it follows the same direction under an adaptive time parameterization. With only  $C^2$  regularity, the safe remainder is  $o(\beta^2)$ . The displayed expansion additionally requires small realized displacement, inactive projection, and an exact statewise solve. At large budgets, our experiments diagnose the resulting nonlocal actor-critic regime instead of extrapolating Eq. (17). The end-to-end return sweep, even at small nominal  $T$ , is not a frozen-critic action-level consistency test of this limit.

**Proposition 4** (Local fixed-budget composition). *Let a smooth one-step update in local coordinates satisfy*

$\Psi_h(x) = x + hf(x) + h^2 p(x) + O(h^3)$ . For fixed  $K$  and  $T \rightarrow 0$ ,

$$\begin{aligned} \Psi_{T/K}^K(x_0) &= x_0 + T f_0 \\ &\quad + T^2 \left[ \frac{K-1}{2K} J_0 f_0 + \frac{1}{K} p_0 \right] + O(T^3). \end{aligned} \quad (18)$$

where  $f_0 = f(x_0)$  and  $J_0 = Df(x_0)$ . Coordinate explicit Euler has  $p_0 = 0$ , whereas exact backward Euler has  $p_0 = J_0 f_0$ . Hence both are first-order methods; for each fixed  $K$ , the  $T \rightarrow 0$  expansion has a leading flow-bias coefficient proportional to  $1/K$ .

Proposition 4 explains a possible fixed-critic small-step benefit of substepping. It neither predicts the sign of true-return change nor applies to the projected, saturated, moving-critic branches studied at large  $T$ . The expansion is not a uniform  $K \rightarrow \infty$  result.

## 5 EXPERIMENTS

We evaluate BAR on the nine D4RL locomotion tasks formed by HalfCheetah, Hopper, and Walker2d with medium, medium-replay, and expert datasets (Fu et al., 2020). These tasks directly exercise deterministic continuous control and provide distinct behavior-data regimes for a target-routing stress test. Base implementations follow CORL configurations (Tarasov et al., 2022); scores are D4RL normalized returns. Additional per-task results and diagnostic definitions are in the Supplementary Material.

**Fixed-budget protocol and estimand.** We hold a total nominal budget  $T$  fixed, split it across  $K$  actor hops as  $h = T/K$ , and sweep

$$T \in \{0.05, 0.1, 0.2, 0.4, 0.7, 1.5, 2.5, 4, 7, 10, 12, 14, 17, 20\}.$$

over all nine tasks. Both grids use seeds 0–3, for  $9 \times 14 \times 4 = 504$  runs per depth. Proximal seeds 0–1 and 2–3 were executed on separate hosts and form disjoint replication pairs. The grid is a high-resolution stability phase diagram rather than a final benchmark alone. For  $K = 1$ , Proposition 1 maps  $T$  to the normalized TD3+BC coefficient  $\alpha = 2T$ .

We compare TD3+BC ( $K = 1$ ), proximal BAR with  $K \in \{2, 3, 4\}$  (BAR-Prox), and an action-metric-matched projected linearized target with  $K \in \{2, 3\}$  (BAR-Lin):

$$\begin{aligned} a_k^{\text{tar}} &= \Pi_{\mathcal{A}} \left[ \nu_k + d \frac{T \nabla_a \hat{Q}(s, \nu_k)}{C_{k,t}^L} \right], \\ \nu_1 &= a, \quad \nu_k = \text{sg } \mu_{k-1,t}^+(s) \quad (k \geq 2). \end{aligned} \quad (19)$$

At each hop, BAR-Lin detaches the reference, recomputes the  $\hat{Q}$  gradient and scale, clips the target to the action box, and takes one mean-squared-error optimizer step with the corresponding persistent actor. Its first-hop scale is

$$C_{1,t}^L = \text{sg}\left(\mathbb{E}_{(s,a)\sim\mathcal{D}}|\hat{Q}_t(s,a)| + \epsilon\right),$$

whereas  $C_{k,t}^L = C_{k,t}^P$  for  $k \geq 2$ . Thus the factor  $d$  matches the action metric, but the first-hop normalization point differs from BAR-Prox; the comparison does not match the complete scale schedule. It uses the same first-hop target and final-hop evaluation routing. All conditions match datasets, critic updates, and the  $10^6$ -update horizon; every headline score is the ten-episode D4RL normalized mean at the final checkpoint. Intermediate evaluation cadence varies by run family and does not enter this estimand. A  $K$ -hop condition performs  $K$  actor optimizer steps per delayed update, so depth comparisons are between complete training procedures. The end-to-end intervention changes local coefficient, re-centering frequency, actor compute, normalization trajectory, routing, and optimization history.

We test whether this intervention reduces large-budget low-return outcomes, persists under the linearized realization, separates first-hop exposure from final-policy reach, and exposes critic-validity signatures of failure.

### 5.1 Additional actors broaden the stability envelope

Figure 2 and Table 1 summarize the sweep. The high-budget estimand averages seeds and the seven  $T \geq 4$  budgets within each task, then weights tasks equally. A seed-mean environment–budget cell is descriptively collapsed below score 20. Across proximal  $K = 1, 2, 3, 4$ , the mean rises  $25.38 \rightarrow 41.20 \rightarrow 51.92 \rightarrow 63.97$  while collapsed cells fall  $34 \rightarrow 22 \rightarrow 16 \rightarrow 8$  of 63.  $K = 4$  gains 12.06 over  $K = 3$  with task-resampling interval  $[3.84, 21.11]$  and seven of nine positive task means, yet its cellwise median is only +.34: eight rescues, no reversals, contribute 54.7% of the gain. The  $T \geq 10$  tail remains strictly ordered,  $20.27 \rightarrow 31.01 \rightarrow 42.23 \rightarrow 56.79$ . BAR-Lin  $K = 3$  raises the four-seed high-budget mean from 38.43 to 46.44 and reduces collapsed cells from 25 to 21; its +1.37 cellwise median indicates a broader shift. The largest proximal gain therefore acts chiefly through lower-tail rescue rather than a uniform shift.

**Task dependence and replication.** Responses range from smooth to sharply nonmonotone, so there is no universal critic-valid radius. Nominal  $T$  is the

Table 1: Aggregate fixed-budget results. High denotes  $T \geq 4$  and tail denotes  $T \geq 10$ . Collapse entries are seed-mean cells / raw seeded runs; P and L denote proximal and linearized.

| Method | Seeds | $T \leq 2.5$ | High         | Tail         | Collapse      |
|--------|-------|--------------|--------------|--------------|---------------|
| TD3+BC | 4     | 66.68        | 25.38        | 20.27        | 34 / 142      |
| BAR-P2 | 4     | 70.64        | 41.20        | 31.01        | 22 / 111      |
| BAR-P3 | 4     | 70.53        | 51.92        | 42.23        | 16 / 84       |
| BAR-P4 | 4     | <b>70.82</b> | <b>63.97</b> | <b>56.79</b> | <b>8 / 54</b> |
| BAR-L2 | 4     | 69.79        | 38.43        | 28.47        | 25 / 112      |
| BAR-L3 | 4     | 69.73        | 46.44        | 35.83        | 21 / 95       |

strongest absolute within-environment rank predictor of return in eight of nine environments; realized movement is a mechanism readout, not a substitute for the intervention. P4–P3 is positive within all three dynamics families (+.79, +13.04, +22.33), and every leave-one-family-out mean remains positive (+6.92 to +17.69). Both host-separated seed pairs reproduce the high-budget ordering: 25.97, 42.02, 52.78, 63.61 and 24.79, 40.38, 51.05, 64.33. These are disjoint-seed replications of one implementation.

The linearized sweep shows that the gain is not unique to the proximal loss. At high budgets, BAR-Lin with  $K = 3$  gains 8.01 points over  $K = 2$  (95% task-resampling interval  $[3.56, 13.53]$ ; eight of nine positive task means) and reduces collapse cells from 25 to 21. Both disjoint seed pairs reproduce the direction, with gains of 7.23 and 8.80. BAR-Prox remains stronger overall, so these are comparisons of complete training procedures rather than isolated integrator effects.

### 5.2 Large-budget behavior is nonlocal and critic dependent

At  $T = 20$ , the projected BAR-Lin  $K = 2$  target is clipped at an action bound in 94–95% of states, and the effective action-step coefficient  $h_{\text{eff}}$  is only 14% and 9% of the nominal  $dT/K$  coefficient in the two hops. These are medians over 18 environment–seed runs per hop. The practical operation is projected target generation followed by amortized neural regression under a moving critic. The sweep measures the resulting end-to-end stability envelope; local truncation error lies outside this estimand.

Figure 3 shows the common failure signature. Environment return supplies the external label: collapsed runs score below 20. These runs have orders-of-magnitude larger TD-error tails, larger critic magnitudes, and larger realized movements. The controlled common critic is the frozen final TD3+BC  $T = 1$  critic matched by environment and seed. A run-total gain sums the mean  $Q$  change across actor hops on the

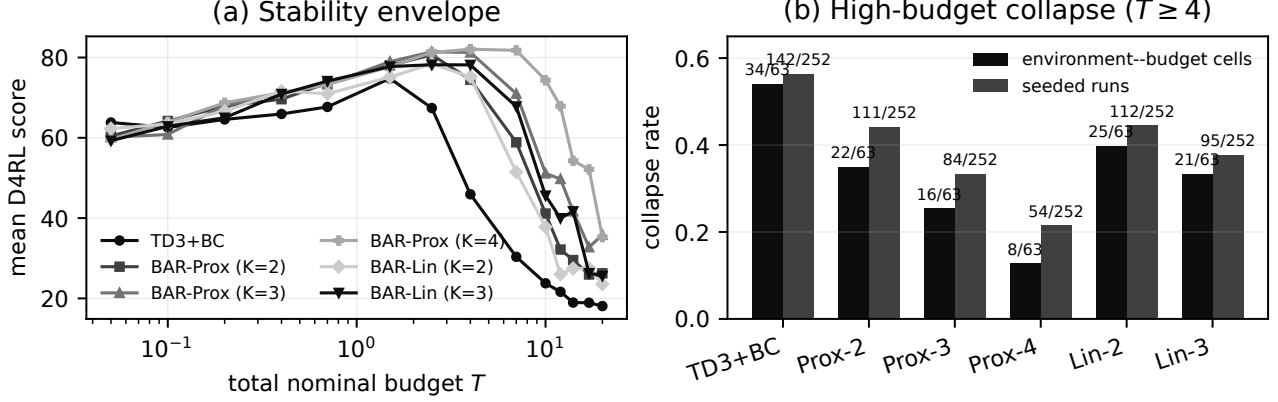


Figure 2: Fixed-budget sweep over nine locomotion tasks. (a) Mean score over tasks. (b) Collapse rates for  $T \geq 4$ , reported both by environment-budget cell and by seeded run. All curves and cell means use four seeds. Both realizations broaden the stable high-budget region with depth.

shared state batch. For collapsed BAR-Lin runs, its median under the jointly trained critic is  $3.33 \times 10^{11}$ ; under the common critic it is  $-1.42$ , compared with  $+2.11$  for stable runs. For BAR-Prox, the common-critic gain changes from  $+2.51$  in stable runs to  $-1.06$  in collapsed runs. This sign reversal diagnoses cross-critic disagreement consistent with Eq. (14). A separate four-environment seed-2/3 sensitivity audit, using a fixed TD3+BC  $T = .05$  reference critic, preserves the split: all 45 stable gains are positive and all 35 collapsed gains are negative (Supplement).

A simulator-reference audit provides targeted evidence of critic miscalibration in the screened checkpoints: the median signed error  $\hat{Q}_1(s, a_K) - Q_{MC}^{\rho_1}(s, a_K)$  is  $+1.70 \times 10^{12}$  in five collapsed checkpoints and  $-41.4$  in 55 stable ones, using the median of checkpoint-level state medians. Episode-cluster intervals exclude zero for critic error at 58 of 60 checkpoints, and restoring the sampled episode clock preserves the error sign at all 60. Its action-ranking subset is sparse, so we use it only as targeted checkpoint-level evidence.

### 5.3 Target-action displacement and final-policy reach

At the same nominal budget, BAR-Prox realizes a smaller current-state first-hop displacement proxy than BAR-Lin once  $T \geq 7$  (the Supplementary Material). At  $T = 20$ , their median  $\hat{D}_{critic}$  values are 0.195 and 0.323. Across 90 matched environment-budget-seed cells at  $T \in \{4, 7, 10, 14, 20\}$ , exactly one of BAR-Prox and BAR-Lin collapses in 11 cases. In all 11, the stable realization has both smaller  $\hat{D}_{critic}$  and smaller  $\hat{D}_{final}$ . These matched cells associate smaller current-state movement with stable trajectories; they do not isolate a causal TD-target route.

We measure Eq. (12) on all 270 audited method checkpoints. The current-state first-branch proxy closely tracks the deterministic target-action metric ( $r = .9999$ ). BAR-Prox  $K = 3$  has lower sample-anchored target-action displacement than TD3+BC in 88 of 90 matched cells (median ratio .658), while the current-state  $\hat{D}_{final}$  proxy is lower in only 36 of 90 (median ratio 1.057). A few large final-policy contractions make the arithmetic mean difference negative despite this typical-cell expansion. Thus audited three-hop BAR exhibits the intended asymmetric movement response rather than merely shrinking both branches. A separate four-seed  $K = 4$  dump lowers target displacement in 89/90 matched seed-0/1 comparisons with TD3+BC (median ratio .568) and 85/90 with  $K = 3$  (.950). It measures first-hop exposure, not the final actor; the separate 24-cell frontier panel supplies the only  $K = 4$  final-policy comparison. Added depth therefore continues to suppress bootstrap exposure without establishing universal final-policy contraction. The P3/P2 direction also holds in 36/40 cells of a four-environment seed-2/3 sensitivity (median ratio .905). Yet exposure alone does not determine return: P3 loses by more than one point in 17 of the original 84 lower-exposure cells (Supplement).

A targeted Hopper-medium audit exhibits both forms of decoupling. Over  $T = 4-10$ ,  $K = 3$  extends final-policy reach at nearly unchanged first-hop movement; near the  $T = 12, 14$  transition it contracts both movements and avoids the low-return outcome. At  $T = 20$ , both depths saturate and fail (Supplement).

### 5.4 Targeted component controls

In eight post-hoc runs, actor-detached shadow critics use either first- or final-route target actions.

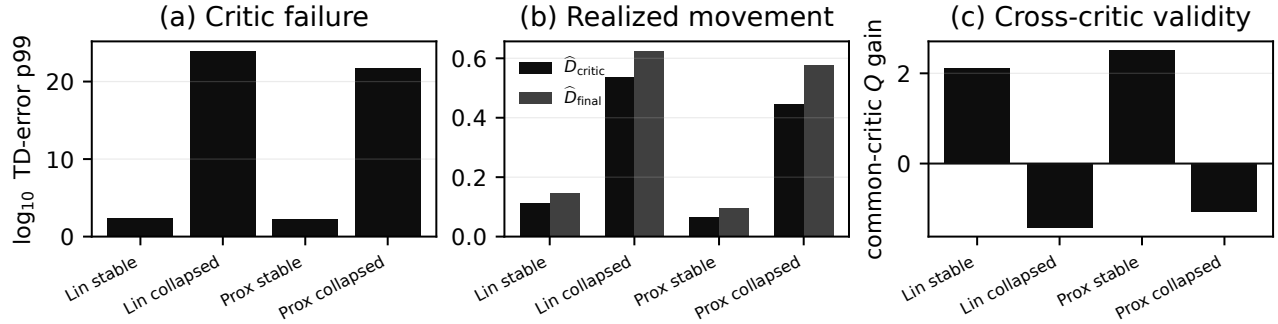


Figure 3: Median diagnostics over the matched  $T \in \{4, 7, 10, 14, 20\}$  audit. Proximal and linearized realizations share a diagnostic pattern consistent with critic miscalibration: critic errors and realized movements increase sharply, and run-total gains under a common critic become negative. Environment return labels failure; the common critic diagnoses cross-critic disagreement.

Native-continuation calibration favors the first route, whereas the common-continuation comparison misses its pre-specified rule; neither estimates a return effect. On the same eight selected environment-budget-seed cells, compute-matched sequential, fixed-reference, and direct-final controls score 71.4, 70.0, and 59.4; slower  $K = 1$  target averaging does not rescue their low returns.

A broader two-seed control removes sequential re-centering while retaining separate target and deployment actors. Across 90 matched high-budget cells, it scores 59.22 versus 57.67 for a contemporaneous BAR-P3 rerun and matches its seed-mean collapse count at 9/45. Control minus BAR is +1.56 with task-resampling interval  $[-1.52, 5.44]$ , while its paired median is  $-0.42$  and BAR wins 52/90 pairs; only 2/9 task means favor the control. Sequential re-centering is therefore not a prerequisite for these observed aggregate outcomes on this grid. The control also changes actor work, anchoring, and the realized critic trajectory, so it does not isolate which element of the simpler procedure is responsible (Supplement).

### 5.5 Discussion and limitations

Both realizations broaden the high-budget stability envelope, with BAR-Prox stronger. The  $K = 3$  audit exhibits exposure-reach asymmetry, while  $K = 4$  adds target-branch contraction and lower-tail rescue. A projection-free frozen-critic audit supports local explicit-implicit agreement; the live result belongs to a coupled finite-step procedure in which depth changes local step, re-centering, actor compute, normalization, routing, and optimization history.

Both headline score grids use four seeds, and the proximal grid forms two host-separated pairs. The full P3 movement and exposure audit uses seeds 0–1; a

restricted four-environment seed-2/3 exposure audit supplies a sensitivity check, while the P4 target dump uses four seeds. For proximal seeds 2–3, provenance consists of the archived launch configuration and score matrices; exact per-run manifests were not retained, and post-hoc P4 manifests do not reconstruct them. Only a 24-cell panel measures matched P4 final displacement. The tasks span three dynamics families, collapse is descriptive, and controls are post hoc. The 90-pair policy-separation control uses seeds 0–1, is not compute matched, and includes ten Walker2d-expert cells completed through a documented CPU recovery chain (eight resumed and two CPU-only). We provide no offline rule for choosing  $T$  or  $K$ . Actor time and memory scale as  $O(K)$ ; the ideal fixed-critic analysis motivates but does not certify the live chain.

## 6 CONCLUSION

BAR turns refinement depth into a practical stability control by routing the first actor into Bellman targets and deploying the final actor. Across four proximal seeds, the  $T \geq 4$  mean rises from 25.38 at  $K = 1$  to 63.97 at  $K = 4$  as low-return cells fall from 34/63 to 8/63; both disjoint seed pairs reproduce the ordering. Across the four-seed linearized grid,  $K = 3$  also gains 8.01 points over  $K = 2$ , and both disjoint seed pairs reproduce the direction. Direct audits identify moderate-depth exposure-reach decoupling and added-depth first-hop contraction, with lower-tail rescue driving most of the latter gain.  $K$  therefore controls a coupled finite-step procedure shaping bootstrap exposure and deployment movement, not merely a discretization-accuracy knob. The two-actor control further shows that sequential re-centering is not a prerequisite for the observed P3 mean/collapse regime, while not isolating which element of the sim-

pler target–deployment-separated procedure is responsible.

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## Checklist

- For all models and algorithms presented, check if you include:
  - A clear description of the mathematical setting, assumptions, algorithm, and/or model. [Yes/No/Not Applicable]  
**Yes.** Sections 3–4 define the setting, actor chain, losses, and assumptions.
  - An analysis of the properties and complexity (time, space, sample size) of any algorithm. [Yes/No/Not Applicable]  
**Yes.** Section 4 states that BAR uses  $K$  persistent actors and  $K$  actor optimizer calls per delayed update; the theoretical properties are stated in Propositions 2–4.
  - (Optional) Anonymized source code, with specification of all dependencies, including external libraries. [Yes/No/Not Applicable]  
**Partial.** The anonymized release contains the BAR launch, evaluation, diagnostic, and verification code plus the compact policy-separation audit. The control-run configs identify local training-source files by SHA-256, but did not record a clean Git revision.
- For any theoretical claim, check if you include:
  - Statements of the full set of assumptions of all theoretical results. [Yes/No/Not Applicable]  
**Yes.** Each proposition states its assumptions, and the local-flow assumptions are separated from the coupled large-budget regime.
  - Complete proofs of all theoretical results. [Yes/No/Not Applicable]  
**Yes.** Complete proofs are provided in the supplementary material.

- (c) Clear explanations of any assumptions. [Yes/No/Not Applicable]  
**Yes.** Sections 4 and 5 explain the learned-critic, fixed- $K$ , small-step, and implementation restrictions.
3. For all figures and tables that present empirical results, check if you include:
- (a) The code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL). [Yes/No/Not Applicable]  
**Yes.** Released CSVs, figure-generation code, a numerical claim verifier, and build instructions reproduce every main figure and aggregate table.
- (b) All the training details (e.g., data splits, hyperparameters, how they were chosen). [Yes/No/Not Applicable]  
**No.** All recoverable settings are listed, but exact per-run BAR-P4 configurations and manifests for proximal seeds 2–3 are unavailable; the policy-separation configs also omit a source revision and environment lock.
- (c) A clear definition of the specific measure or statistics and error bars (e.g., with respect to the random seed after running experiments multiple times). [Yes/No/Not Applicable]  
**Yes.** Captions and text define the equally task-and-budget-weighted high-budget estimand, seed-mean cells, raw seeded runs, collapse thresholds, paired ratios, and task-resampling intervals. Both headline grids use four seeds; proximal seeds form two host-separated pairs. Mechanism audits use the explicitly reported two- or four-seed subsets.
- (d) A description of the computing infrastructure used. (e.g., type of GPUs, internal cluster, or cloud provider). [Yes/No/Not Applicable]  
**No.** The released artifact records software and execution protocols, including the policy-separation CPU recovery chain, but does not contain a complete, independently verified inventory of all external-machine hardware.
4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets, check if you include:
- (a) Citations of the creator If your work uses existing assets. [Yes/No/Not Applicable]  
**Yes.** D4RL, TD3+BC, and the CORL implementation basis are cited.
- (b) The license information of the assets, if applicable. [Yes/No/Not Applicable]  
**Yes.** The code release identifies its license; upstream dependencies retain their own licenses.
- (c) New assets either in the supplemental material or as a URL, if applicable. [Yes/No/Not Applicable]  
**Yes.** An anonymized code and compact-results artifact accompanies the submission.
- (d) Information about consent from data providers/curators. [Yes/No/Not Applicable]  
**Not Applicable.** We use public simulated-control benchmark datasets and collect no human data.
- (e) Discussion of sensible content if applicable, e.g., personally identifiable information or offensive content. [Yes/No/Not Applicable]  
**Not Applicable.** The assets contain simulated locomotion trajectories and no personal or sensitive content.
5. If you used crowdsourcing or conducted research with human subjects, check if you include:
- (a) The full text of instructions given to participants and screenshots. [Yes/No/Not Applicable]  
**Not Applicable.**
- (b) Descriptions of potential participant risks, with links to Institutional Review Board (IRB) approvals if applicable. [Yes/No/Not Applicable]  
**Not Applicable.**
- (c) The estimated hourly wage paid to participants and the total amount spent on participant compensation. [Yes/No/Not Applicable]  
**Not Applicable.**